

ME4150: Airframe Structures and Design

Semester: 8

Experimental and Numerical Stress Analysis of a Rectangular Wing Structure

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1 INTRODUCTION

The wing is the main lifting structure of the aircraft. It is subjected to various loading conditions in-flight. The structural integrity of wing is, therefore, is a critical concern in aerospace industry. To make structural design decisions, the load behaviour of the wing structure in presence of loading should be considered. The contribution of the individual structural members of the wing structure respond differently to the loads.

Normally, wing has primary load bearing structures such as spars, skin panels, and ribs. The loads are transferred between these structural members. Landing gears apply a concentrated load on the component it was fixed. If the LGs are fixed at the wingtip through spar ends, the deformation and the stress may differ significantly with the point of action. This variation happens because of the bending, torsional stiffness and the shear flow distribution between spars.

This report depicts an experimental and numerical investigation of the structural behaviour of a rectangular wing structure under two loading conditions.

- 1- Load applied to the front spar tip
- 2- Load applied to the rear spar tip

The experimental work was carried out using TrueStructures Strain Measuring Instrument [1] while Finite Element Analysis was carried out using ANSYS Static Structural to simulate both cases. The main aim is to identify the convenient location for the LG attachment based on the load distribution.

2 EXPERIMENTAL ANALYSIS

2.1 Experimental Setup



Figure 1: TrueStructures setup

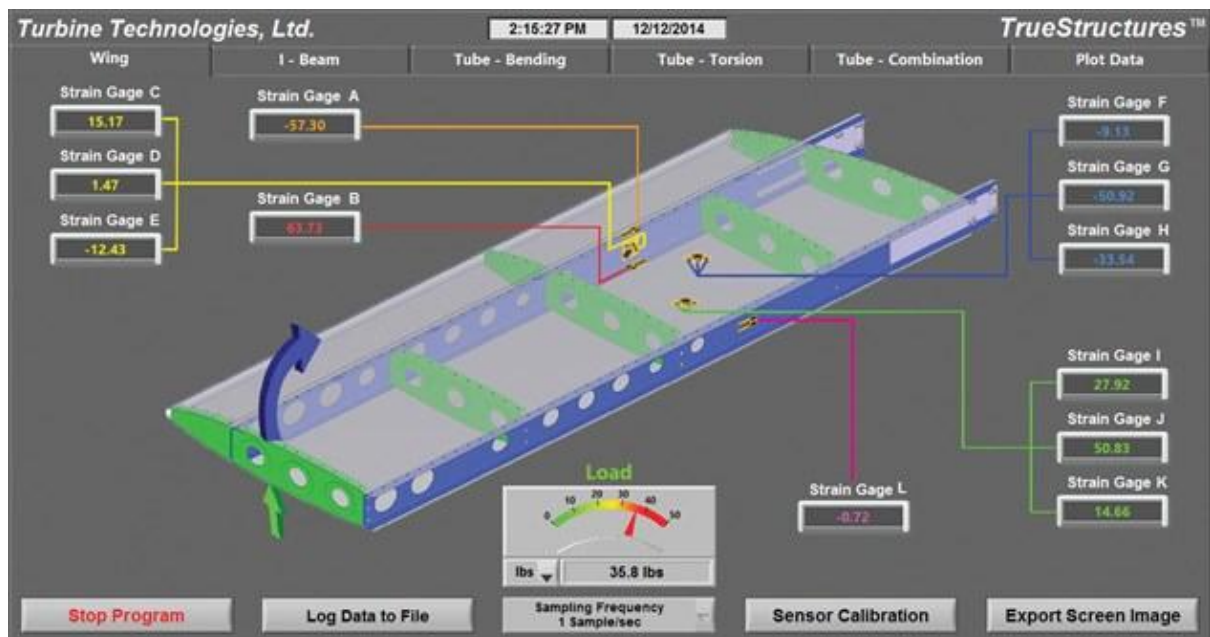


Figure 2: Placement of strain gauges

2.2 Procedures

- The wing prototype was mounted in the rig with root firmly clamped and the bridge controller connector was plugged to the wing.
- Loading wheel was rotated to remove the loads.
- The unloaded load cell readings were recorded from Channel A to Channel L.
- Then the wing was moved as the load applied location is set with the front spar.
- The loading wheel was rotated to applied with 30 lbs.
- Then the load cell readings were recorded from Channel A to Channel L.
- The wing is again moved until the rear spar location with set with the load applied location.
- The load cell readings were recorded from Channel A to Channel L.
- Using the Strain Calculator, the strain at each location for both cases were calculated and recorded.

Strain gauges are placed as below:

- A and B is attached to the upper and lower flange of the front spar.
- C, D and E tri-axial rosette is attached to the web of the front spar.
- F, G, and H tri-axial rosette is attached to the upper skin.
- I, J, and K tri-axial rosette is attached to the lower skin.
- L is attached to the web of the rear spar.



Figure 3: Tri-axial rosette gauge placement

Uni axial gauge point stresses are directly calculated as follows:

Applied Load = 30lb

Excitation Voltage(mV) = 15000

Gauge Factor = 2.09

Young's Modulus = 69 GPa

$$\sigma = E\epsilon$$

Rosette's gauge stresses are calculated as taking each rosette's gauge order as listed (first gauge = 0°, middle = 45°, last = 90°) for the Mohr's Circle construction. In strain gauge rosette, at least three strain measurements taken in different directions at the same point are required. In this experiment, each rosette location (C-D-E, F-G-H, and I-J-K) uses a rectangular (0°/45°/90°) rosette, consisting of three individual gauges arranged 45° apart. The single gauges (A, B, and L) measure strain along one direction only and are treated as uniaxial stress points.

Strain Transformation Equation

The normal strain measured by a gauge oriented at angle θ from a reference x-axis is related to the strain state ($\epsilon_x, \epsilon_y, \gamma_{xy}$) by the strain transformation equation:

$$\epsilon_{\theta} = \epsilon_x \cos^2 \theta + \epsilon_y \sin^2 \theta + \gamma_{xy} \sin \theta \cos \theta$$

For a rectangular rosette with gauges at 0°, 45°, and 90°, applying this equation at each angle and solving simultaneously gives:

$$\epsilon_x = \epsilon_{0^\circ}$$

$$\epsilon_y = \epsilon_{90^\circ}$$

$$\gamma_{xy} = 2\epsilon_{45^\circ} - \epsilon_{0^\circ} - \epsilon_{90^\circ}$$

Mohr's Circle provides a graphical (and equivalently, closed-form algebraic) method for determining principal strains and their orientation from a known strain state.

Construction principle: Plotting normal strain ϵ on the horizontal axis and half the shear strain $\gamma/2$ on the vertical axis, the strain states at a point is represented by two diametrically opposite points:

- Point X = ($\epsilon_x, +\gamma_{xy}/2$)
- Point Y = ($\epsilon_y, -\gamma_{xy}/2$)

A circle is drawn through these two points, centered on the horizontal axis at:

$$C = \epsilon_{avg} = \frac{\epsilon_x + \epsilon_y}{2}$$

with radius:

$$R = \sqrt{\left(\frac{\epsilon_x - \epsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}$$

Principal strains: The points where the circle intersect the horizontal axis (where shear strain is zero) give the principal strains:

$$\varepsilon_1 = \varepsilon_{avg} + R(\text{maximum principal strain})$$

$$\varepsilon_2 = \varepsilon_{avg} - R(\text{minimum principal strain})$$

Principal angle: The orientation of the principal strain axes relative to the 0° reference gauge is:

$$\theta_p = \frac{1}{2} \tan^{-1} \left(\frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y} \right)$$

This angle represents the physical rotation, on the specimen, from the 0° gauge direction to the nearest principal strain axis.

Since a rosette is mounted on a free surface, the point is in a state of plane stress (zero stress normal to the surface). For an isotropic, linear-elastic material, the principal stresses are related to the principal strains through the plane-stress form of Hooke's Law:

$$\sigma_1 = \frac{E}{1 - \nu^2} (\varepsilon_1 + \nu \varepsilon_2)$$

$$\sigma_2 = \frac{E}{1 - \nu^2} (\varepsilon_2 + \nu \varepsilon_1)$$

where:

- E = Young's modulus of the material
- ν = Poisson's ratio of the material
- $\varepsilon_1, \varepsilon_2$ = principal strains obtained from Mohr's Circle

The maximum in-plane shear stress at the point is:

$$\tau_{max} = \frac{\sigma_1 - \sigma_2}{2}$$

The principal stress directions are coincident with the principal strain directions (θ_p), since the material is assumed isotropic.

2.3 Results and Analysis

Load on Front Spar

Table 1: Uni axial strain and stress values when load is applied to front spar

Gauge	Unloaded Voltage (mV)	Loaded Voltage (mV)	Strain ($\mu\epsilon$)	Stress (MPa)
A	228.8	221	-994.18	-68.5984
B	300	307.9	1009.037	69.62355
L	349.3	348.5	-102.06	-7.04214

Table 2: Tri-axial rosette strain and stress values when load is applied to front spar

Rosette	ϵ_1 ($\mu\epsilon$)	ϵ_2 ($\mu\epsilon$)	θ_p	σ_1 (MPa)	σ_2 (MPa)	$\tau_{\max}$ (MPa)
C-D-E	153.6	-179.1	2.2°	7.32	-9.94	8.63
F-G-H	185.3	-1065.0	-30.0°	-12.87	-77.73	32.43
I-J-K	812.7	-187.2	30.9°	58.15	6.27	25.94

The maximum stress value is visualized at strain gauge B where it's attached to the lower flange of the front spar.

Load on Rear Spar

Table 3: Uni axial strain and stress values when load is applied to rear spar

Gauge	Unloaded Voltage (mV)	Loaded Voltage (mV)	Strain ($\mu\epsilon$)	Stress (MPa)
A	229.4	226	-433.61	-29.9191
B	300.9	307.5	842.847	58.15644
L	349.6	350	510.39	35.21691

Table 4: Tri-axial rosette strain and stress values when load is applied to rear spar

Rosette	ϵ_1 ($\mu\epsilon$)	ϵ_2 ($\mu\epsilon$)	θ_p	σ_1 (MPa)	σ_2 (MPa)	$\tau_{\max}$ (MPa)
C-D-E	217	-38.3	0.0°	15.82	2.58	6.62
F-G-H	125.9	-495.9	-52.7°	-2.92	-35.18	16.13
I-J-K	664.1	-128.0	55.4°	48.15	7.06	20.55

The maximum stress value is visualized at strain gauge B where it's attached to the lower flange of the front spar.

Experimental Stress and Strain Distribution

The distribution of stress along the strain gauge locations showed that the load path was changed when the point load was moved from the front spar to the rear spar. In the front spar loading case, large opposite stresses were measured at gauges A and B, which are attached to the upper and lower flange of the front spar. Large stress values were also obtained in the upper and lower skin gauges. This indicates that the front spar and the skin panels carried a large part of the bending and torsional response. In the rear spar loading case, the stress at gauge L, which is attached to the rear spar web, increased clearly. However, the maximum measured stress of the structure is reduced compared with the front spar loading case.

Table 5: Comparison of uni-axial gauge stresses in two cases

Gauge	Stress when load is applied to front spar (MPa) – Case 01	Stress when load is applied to rear spar (MPa) – Case 02	Change in stress of case 02 relative to case 01 (%)
A	-68.5984	-29.9191	56.39
B	69.62355	58.15644	16.47
L	-7.04214	35.21691	-400.09

Table 6: Comparison of tri-axial rosette stresses in two cases

Gauge	Stress when load is applied to front spar (MPa) – Case 01	Stress when load is applied to rear spar (MPa) – Case 02	Change in stress of case 02 relative to case 01 (%)
Principal stress 01			
C-D-E	7.32	15.82	116.1202
F-G-H	-12.87	-2.92	-77.3116
I-J-K	58.15	48.15	-17.1969
Principal stress 02			
C-D-E	-9.94	2.58	-125.956
F-G-H	-77.73	-35.18	-54.7408
I-J-K	6.27	7.06	12.59968
Shear stress			
C-D-E	8.63	6.62	-23.2908
F-G-H	32.43	16.13	-50.2621
I-J-K	25.94	20.55	-20.7787

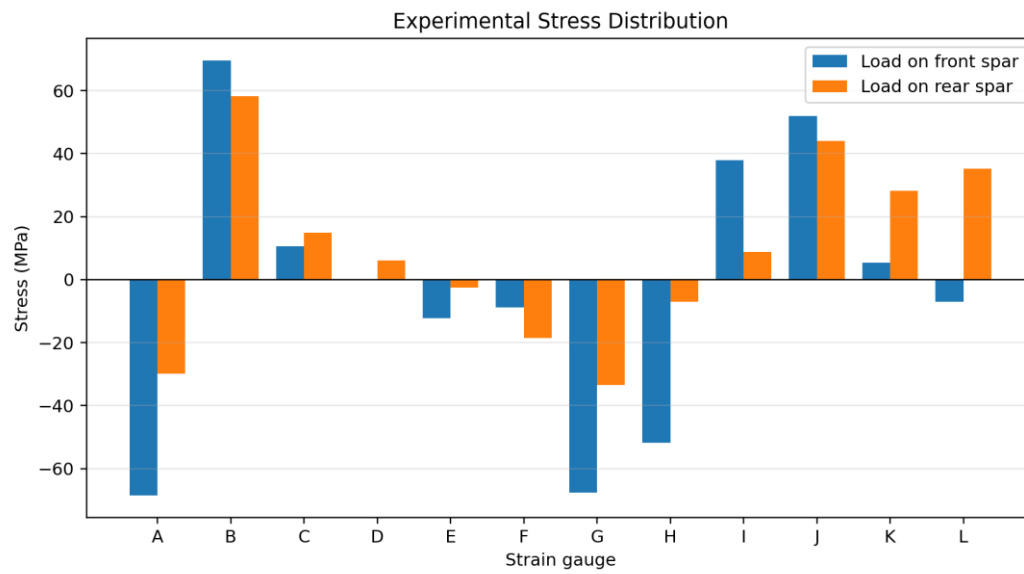


Figure 4: Experimental Stress comparison of front spar and rear spar

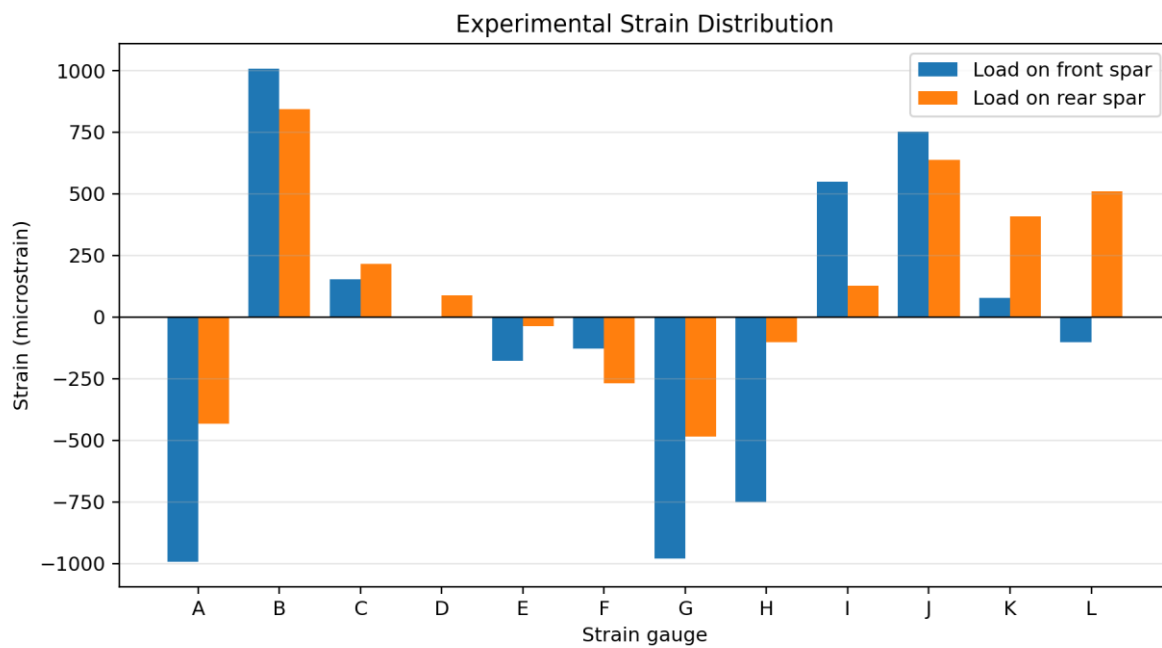


Figure 5: Experimental Strain comparison of front spar and rear spar

3 NUMERICAL ANALYSIS

3.1 Finite Element Setup

The numerical analysis was carried out under same loading and boundary conditions to predict the structural response of the wing structure due to the applied load. The finite element model was developed using the CAD geometry created in SolidWorks and analysed in ANSYS Static Structural.

The main goal of the numerical model was to obtain the strain distribution of the wing structure and compare it with the experimental strain values obtained during the experiment. This comparison was used to validate the accuracy of the finite element model. During this validation, the following procedures were followed.

3.2 Methodology

3.2.1 Wing structure model

Having obtained the dimensions, the wing structure was modelled in Solid Works. For the simplification of the analysis and the lack of available data of the actual structure, Aluminium 6061 T6 alloy was used for the whole structure as the material. The wing structure CAD model includes main structural components, such as the front and rear spars, ribs, skin and the root mounting holes. Briefly, the two main spars were modelled initially. The outer shape of the aerofoil was determined using an image of the side view of the wing structure and modelled using splines. The shape was extruded as a surface and thickened to the desired thickness of the outer skin. Following that, the perpendicular ribs and the angled rib were added. Finally, the structural details such as lightening holes, flange thicknesses and mounting holes were added.

Apart from the actual wing structure model, another model with some geometrical simplifications was created to reduce the computational time and improve the mesh quality. Rivets, rivet holes, small edges, small fillets and non-structural details were neglected since they don't significantly affect the global stiffness of the wing structure. These simplifications helped to avoid poor-quality mesh elements during the meshing phase.

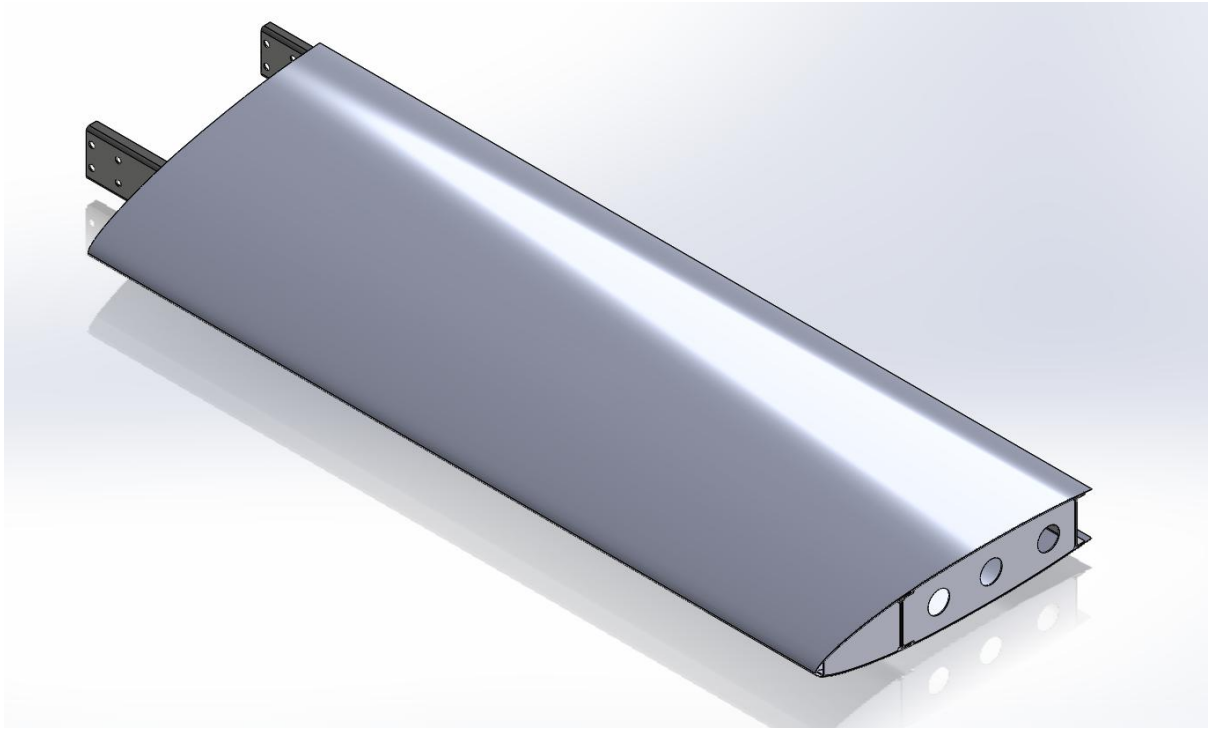


Figure 6: Wing model

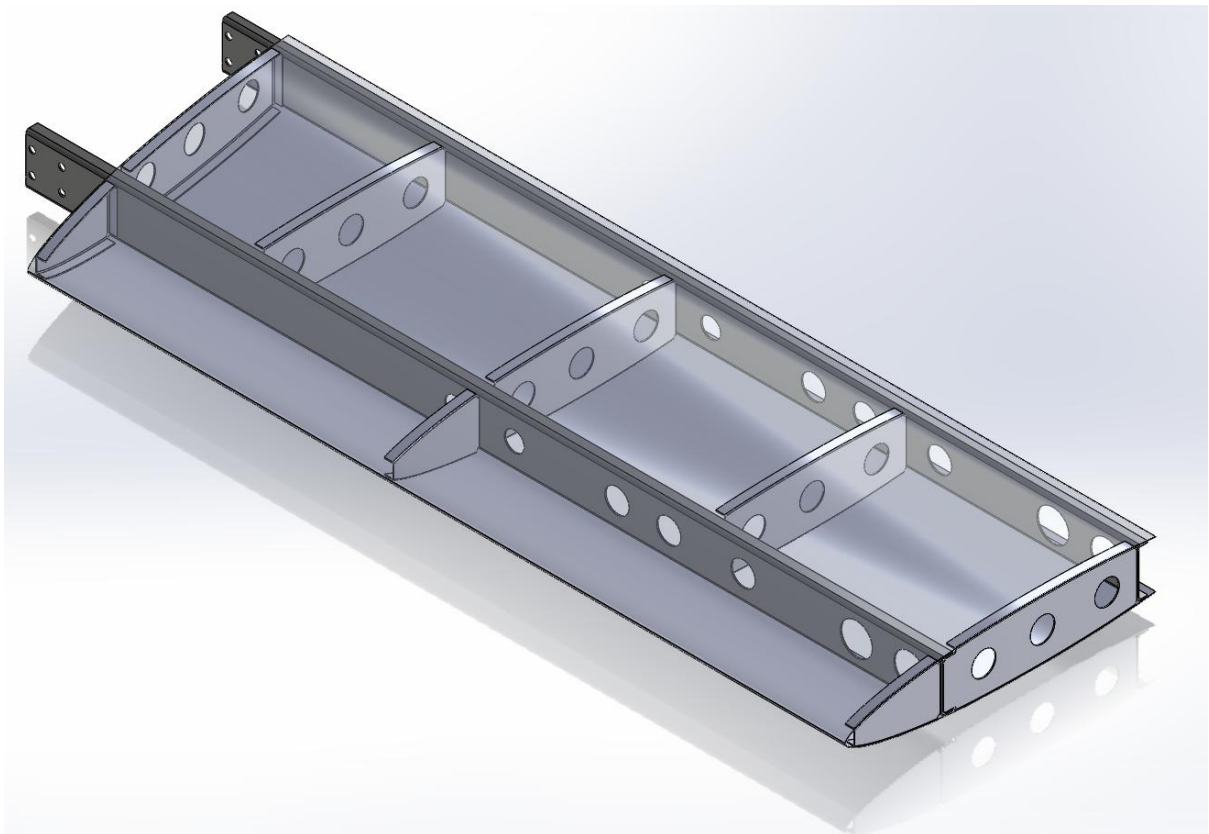


Figure 7: Wing model (Inside structure)

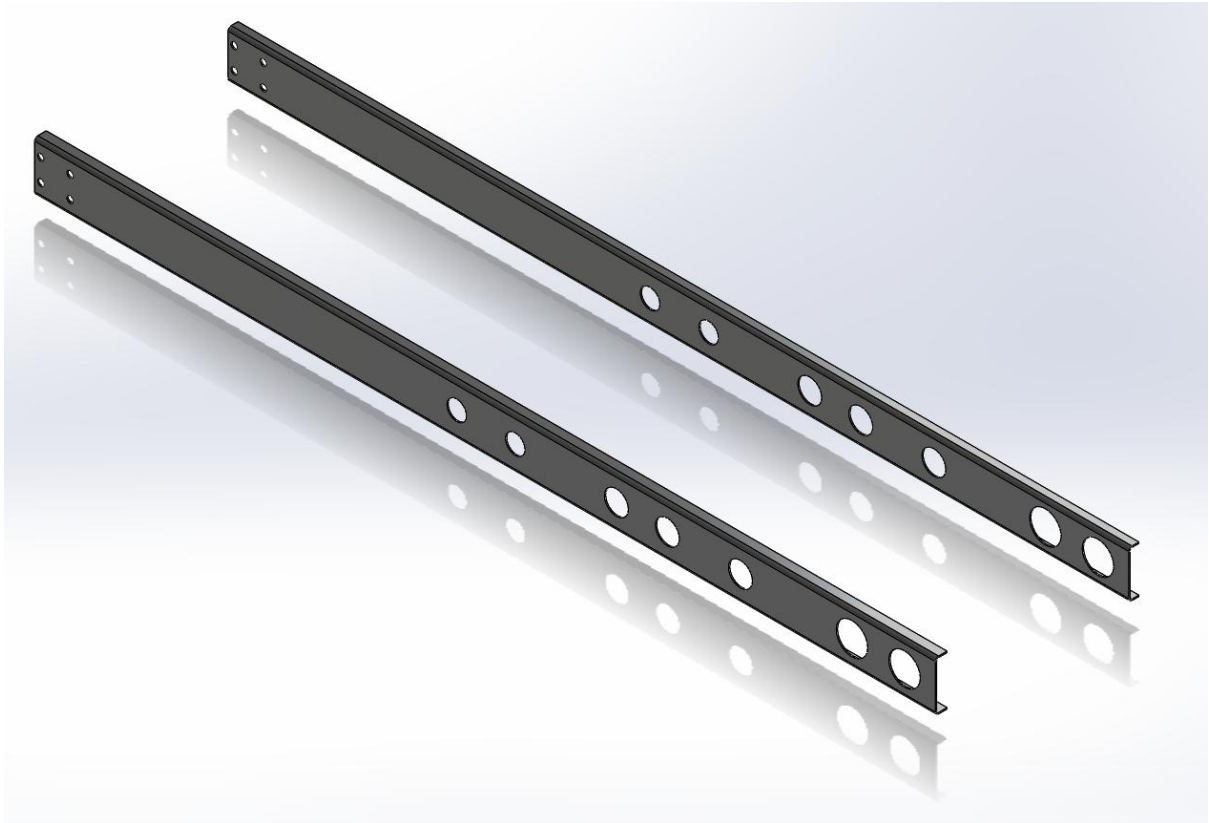


Figure 8: Front and rear spar

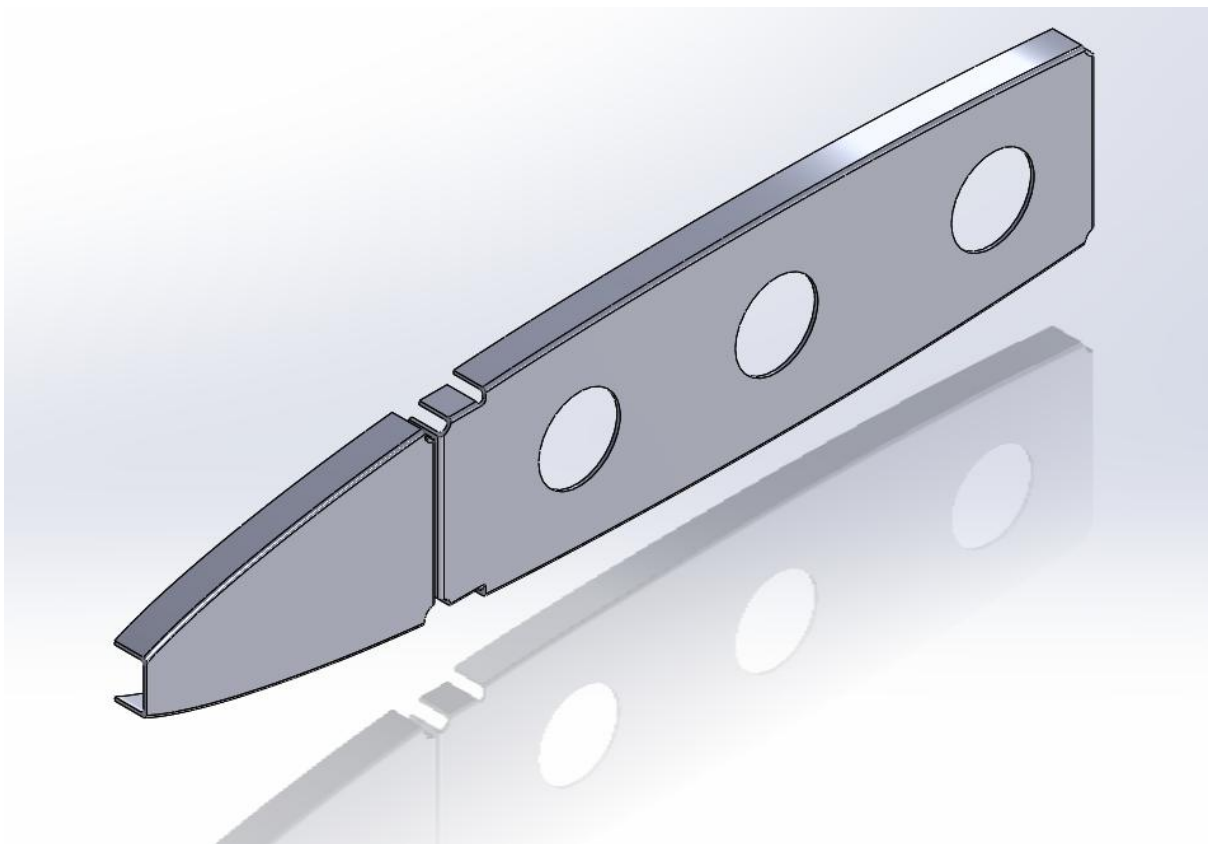


Figure 9: Rib section

3.2.2 Mesh independence test

Tetrahedral elements were used for meshing the model since the wing structure contains several complex geometric features, including lightening holes, curved spar edges, spar-rib intersections, and local cut outs for stress reliefs. Compared with hexahedral elements, tetrahedral elements are easier to generate for complex assemblies and irregular regions for these types of geometries. Creating a fully structured hexahedral mesh for the wing would require significant geometry partitioning and manual control, which would increase modelling time and may introduce meshing difficulties. Therefore, tetrahedral meshing was selected as a practical and efficient approach for this validation study.

A mesh independence test was carried out to ensure that the finite element results were not significantly affected by the selected mesh size. In finite element analysis, the geometry is divided into small elements. If the mesh has larger mesh elements, the results may not be accurate because the deformation and strain distribution cannot be captured properly. However, if the mesh has small mesh elements, the simulation requires more computational time. Therefore, a mesh independence test was performed to select a suitable mesh that gives accurate results with reasonable computational cost.

In this analysis, separate mesh body sizing controls were applied to the ribs, spars, and skin of the wing structure. The experiment had two main cases where 30lb point load is applied at the tip of front and rear spar. Since the loading condition doesn't change the mesh elements structure of the wing model, case with 30lb on the front spar was selected for mesh independence analysis by varying the element sizes of ribs, spars and skin section. The results of the mesh independence can be visualized as shown in Table 7, Figure 10, Figure 11 and Figure 12.

Table 7: Mesh independence study

Ribs Element Size [mm]	Spars Element Size [mm]	Skin Element Size [mm]	Total Deformation Maximum [m]	Equivalent Elastic Strain Maximum	Equivalent Stress Maximum [MPa]	Mesh Elements
8	8	11	0.002381123	0.001058508	73.079	136838
7.8	7.8	10.8	0.002399928	0.001076764	74.340	142408
7.6	7.6	10.6	0.002388891	0.001050103	72.499	148075
7.4	7.4	10.4	0.002407246	0.00105441	72.796	152574
7.2	7.2	10.2	0.00241687	0.001055989	72.902	158490
7	7	10	0.002420651	0.001557912	84.508	163020
6.8	6.8	9.8	0.002421865	0.001081895	74.694	171817

6.6	6.6	9.6	0.002451531	0.001086891	75.038	177949
6.4	6.4	9.4	0.002430616	0.001093891	75.522	185070
6.2	6.2	9.2	0.002448104	0.001047205	72.278	194931

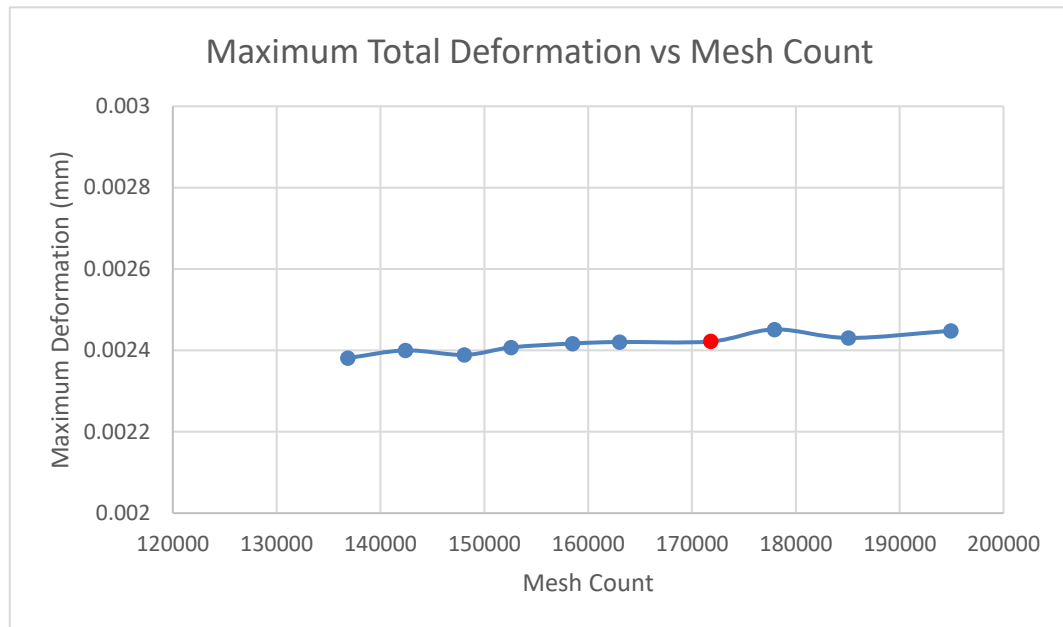


Figure 10: Maximum deformation vs Mesh count

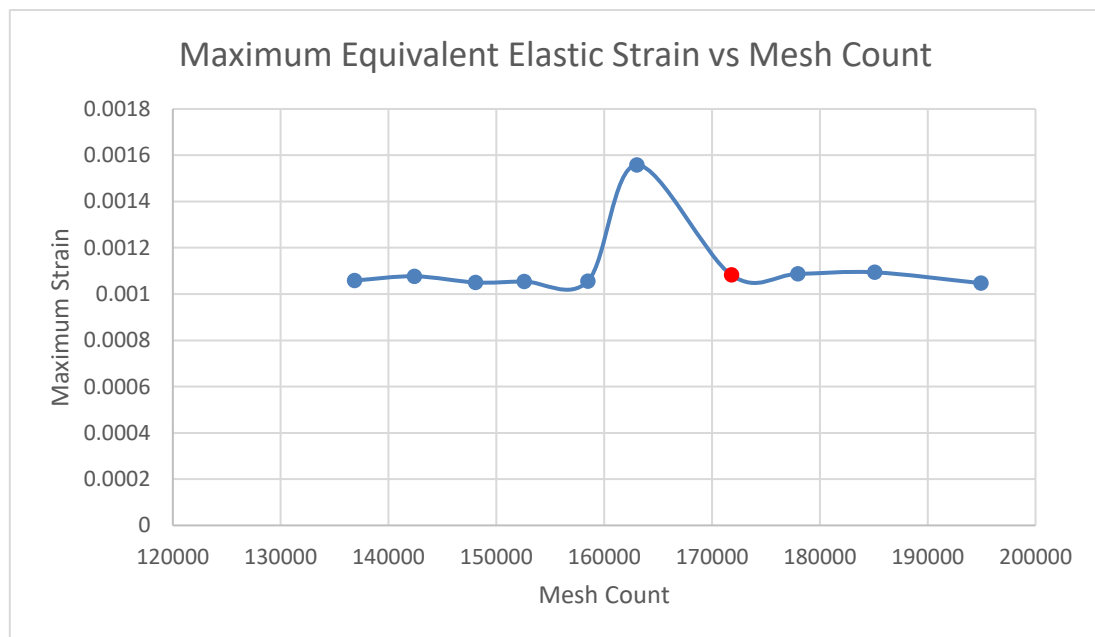


Figure 11: Maximum equivalent elastic strain vs Mesh count

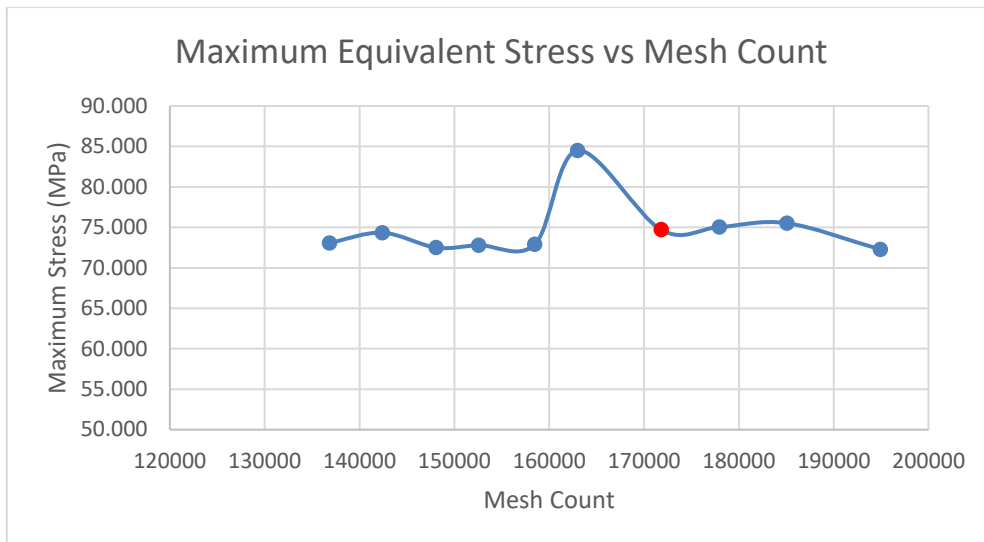


Figure 12: Maximum equivalent stress vs Mesh count

Observing the mesh independence results, the solutions have already been converged up to a certain level. So without further refinement of the mesh, the mesh settings configuration with 171817 mesh elements (As indicated from red dot) was chosen as the converged solution for further studies.

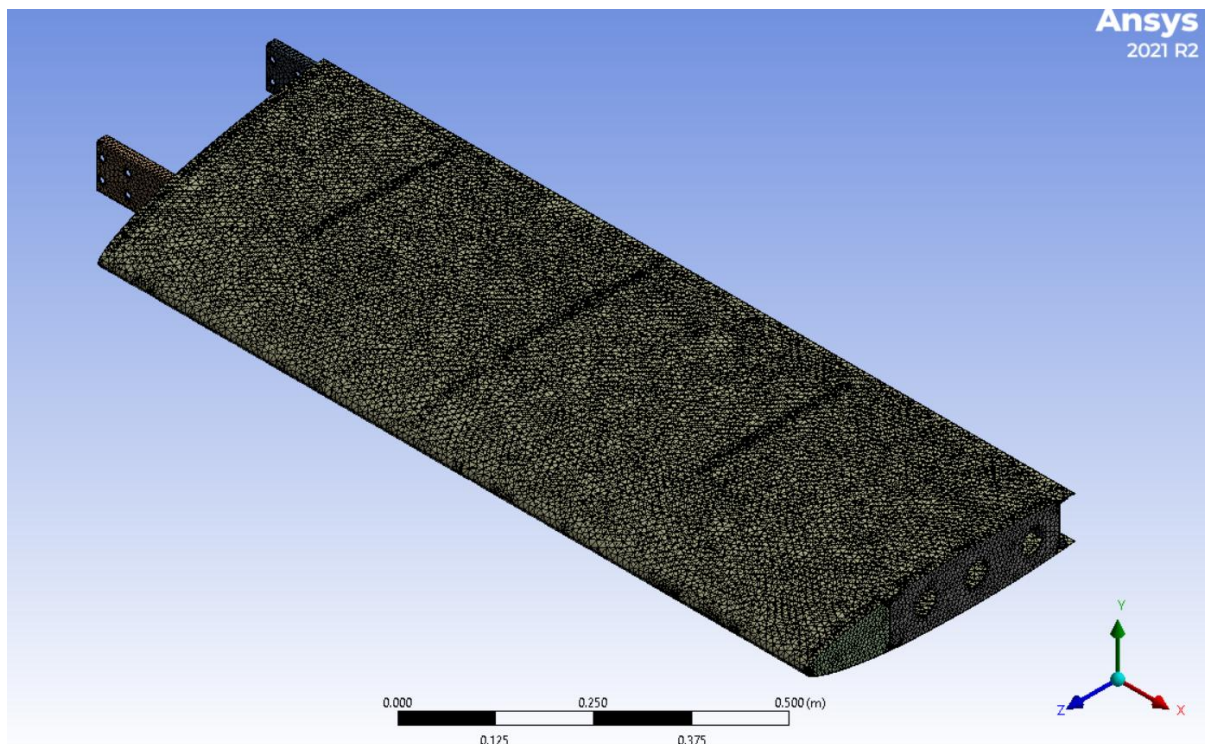


Figure 13: Meshed wing structure

3.2.3 Boundary conditions

Mainly, two boundary conditions were applied. The whole wing structure was held by eight bolts at the root section as shown by the Figure 14. The bolt holes were modelled as fixed faces to represent the actual constraint provided by the bolts in the setup. In the physical wing structure, the bolts restrict movement of the wing root by clamping the structure to the support fixture in blue colour. Therefore, fixing the inner faces of the bolt holes provides a simplified representation of the bolted connection as shown in Figure 15 and Figure 16.



Figure 14: Wing structure mounts

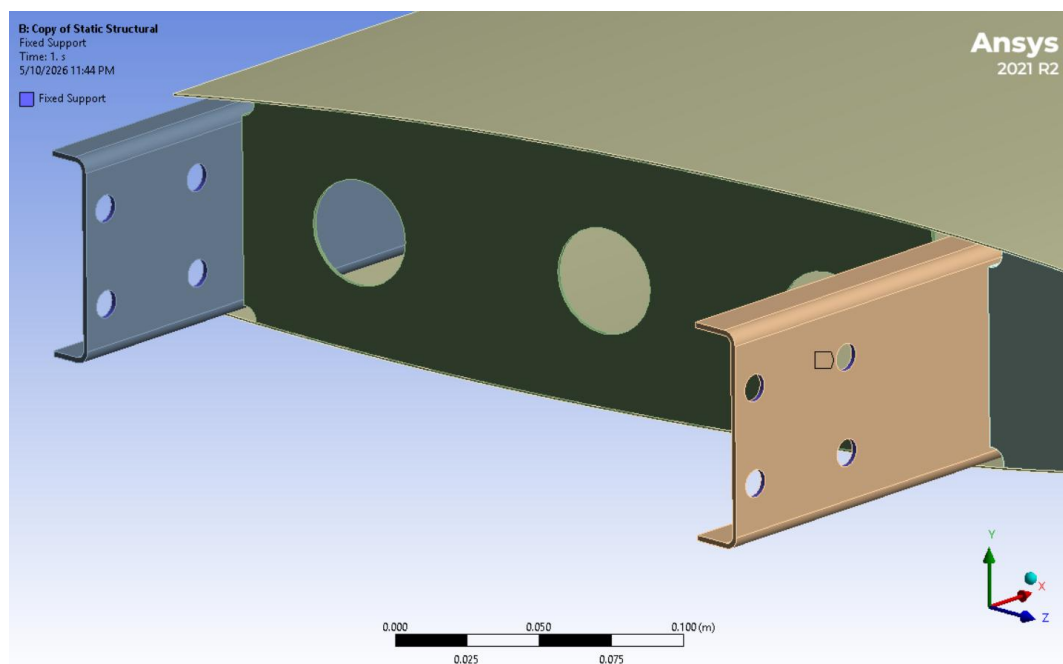


Figure 15: Fixed end condition

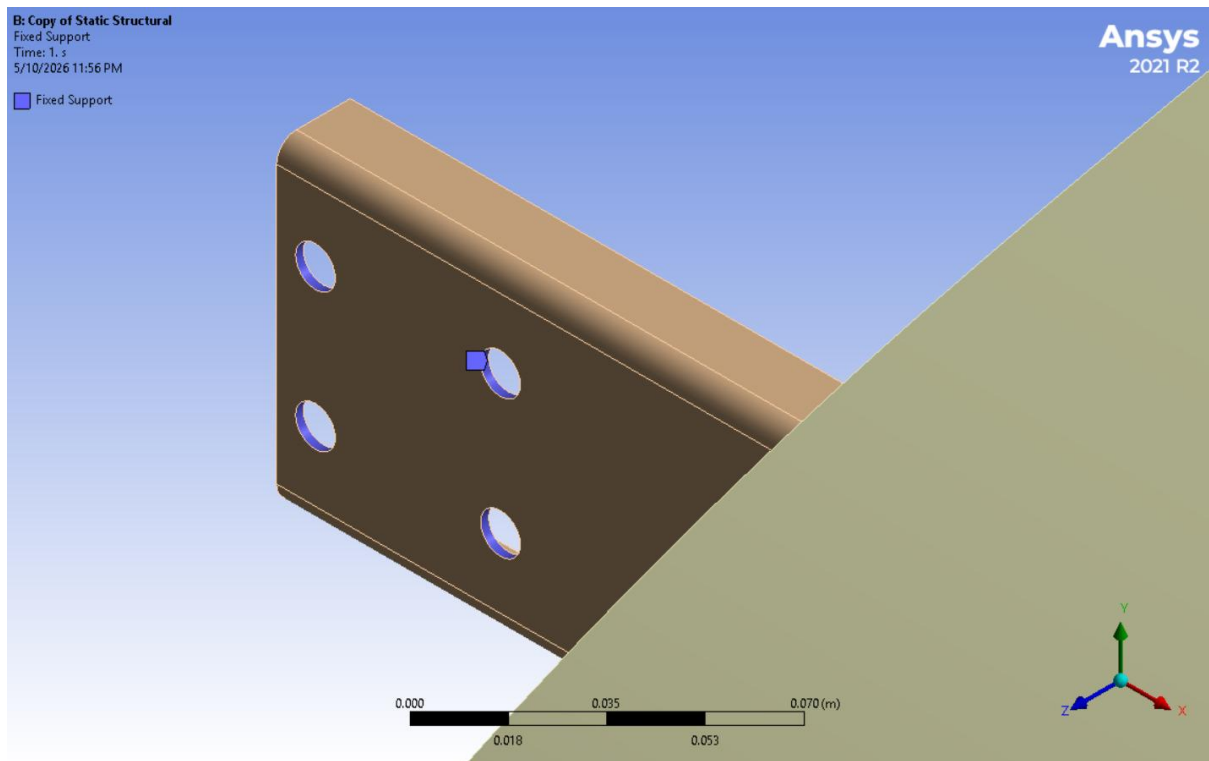


Figure 16: Zoomed in view of the fixed faces

An external point load of 30lb (133.45N) was applied in vertical upward direction to the tip of the front and rear spar of each case as shown in Figure 17.

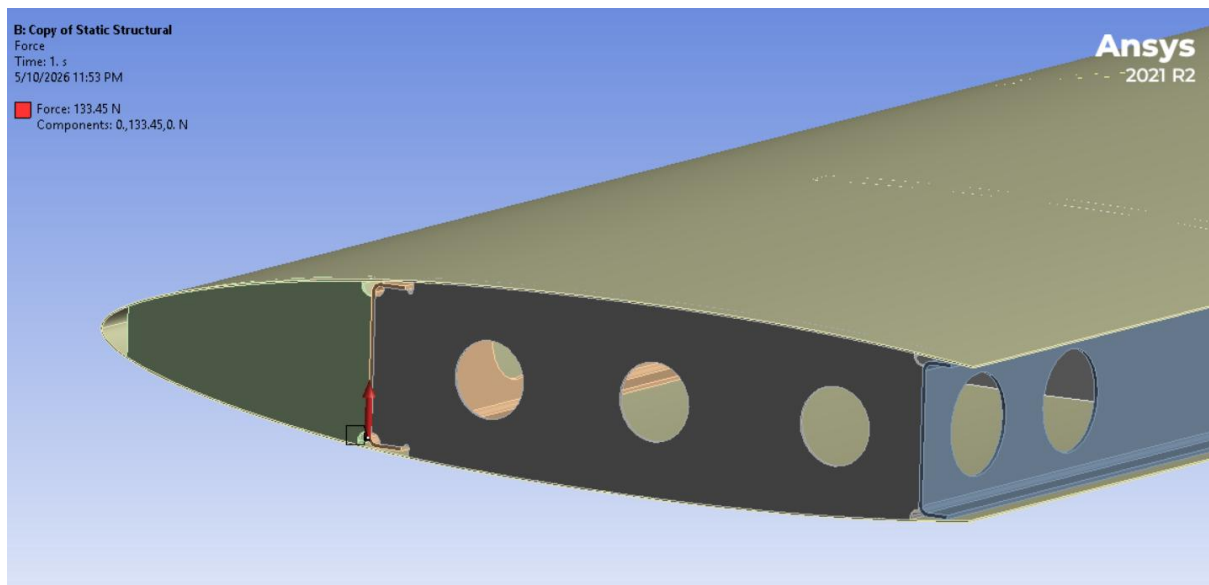


Figure 17: Point load of 30lb on front spar tip

3.3 Results and Analysis

Final results obtained by conducting the finite element analysis for load applied to two spars are listed as follows.

Table 8: FEA Results

	Maximum Deformation (mm)	Maximum equivalent Stress (MPa)	Maximum equivalent Strain
Load on Front spar	2.422	74.694	1.082×10^{-3}
Load on Rear spar	2.491	77.411	1.122×10^{-3}

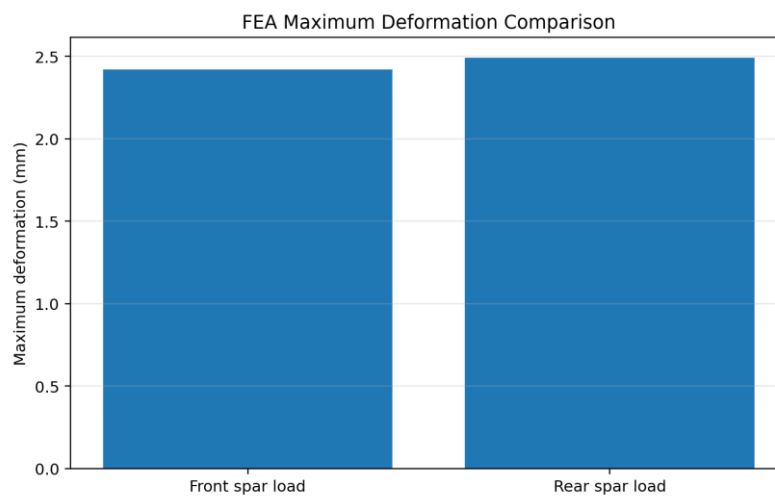


Figure 18 - Comparison of maximum deformation in two cases

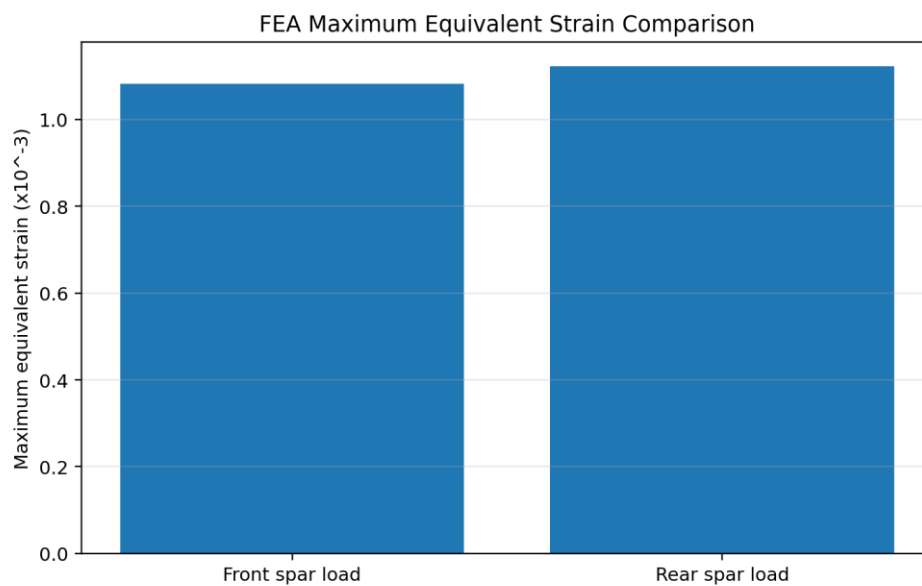


Figure 19: Comparison of maximum equivalent strain for two cases

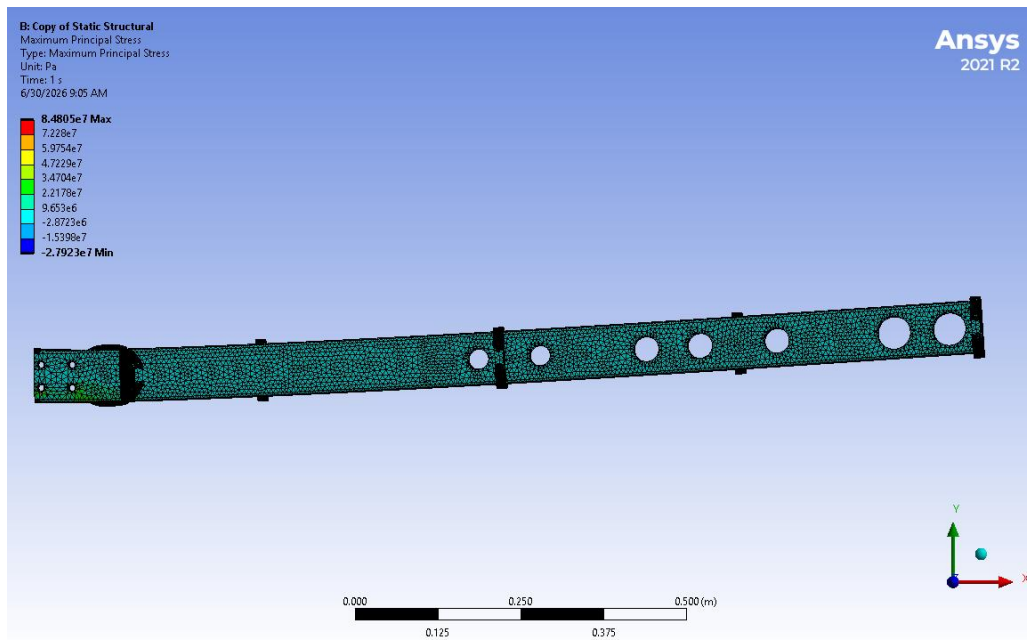


Figure 20: Maximum principal stress variation for front spar loaded case

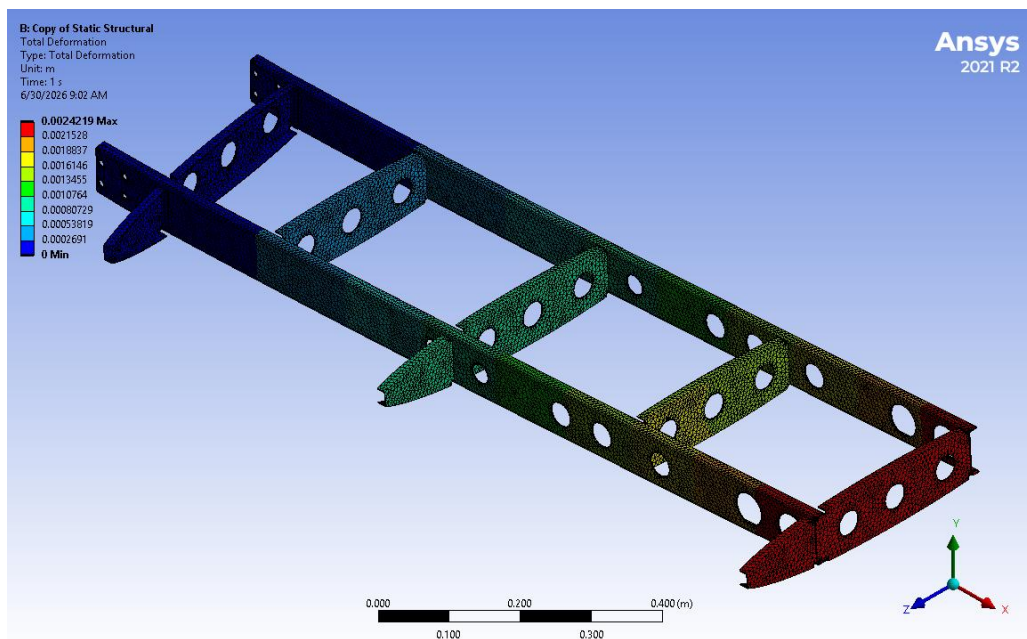


Figure 21: Total deformation variation for front spar loaded case

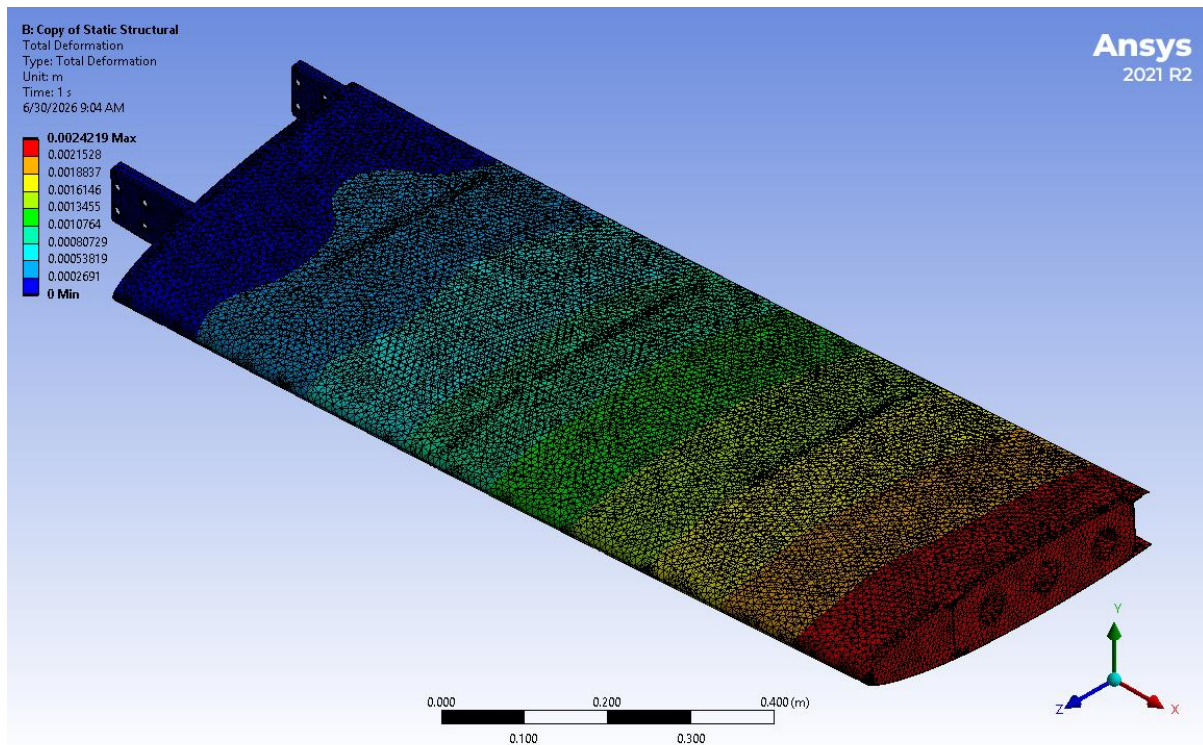


Figure 22: Total deformation variation for front spar loaded case

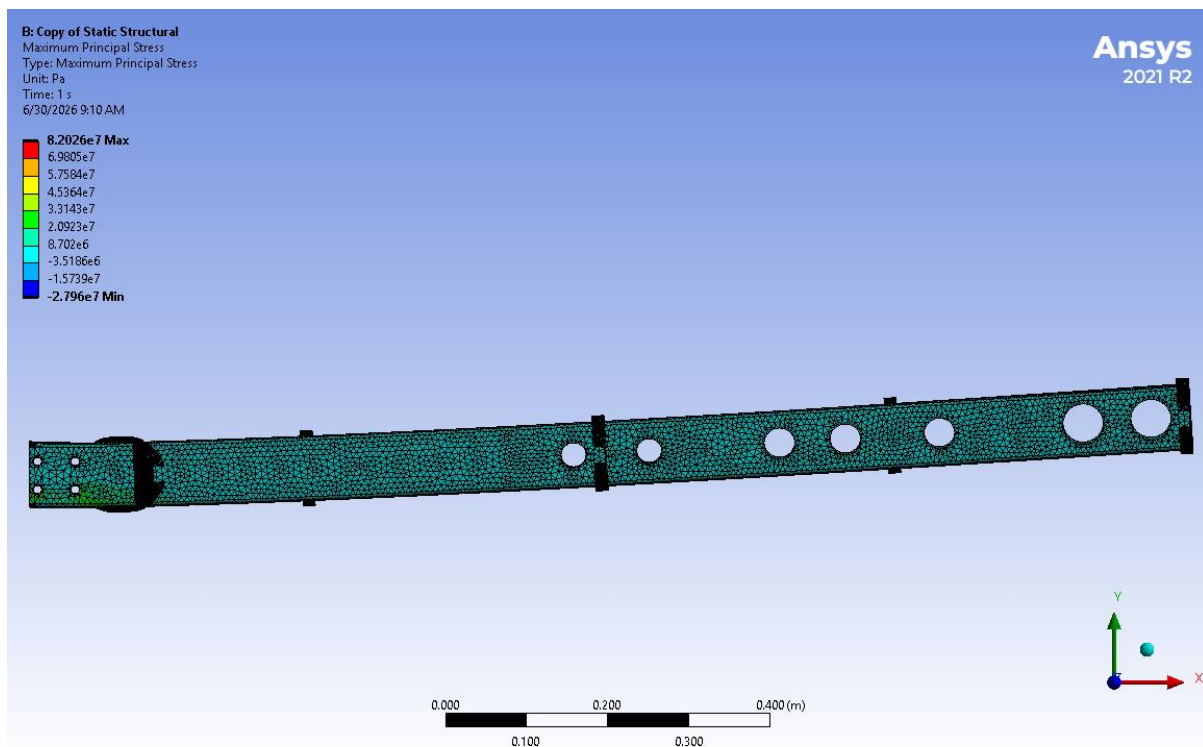


Figure 23: Maximum principal stress variation for rear spar loaded case

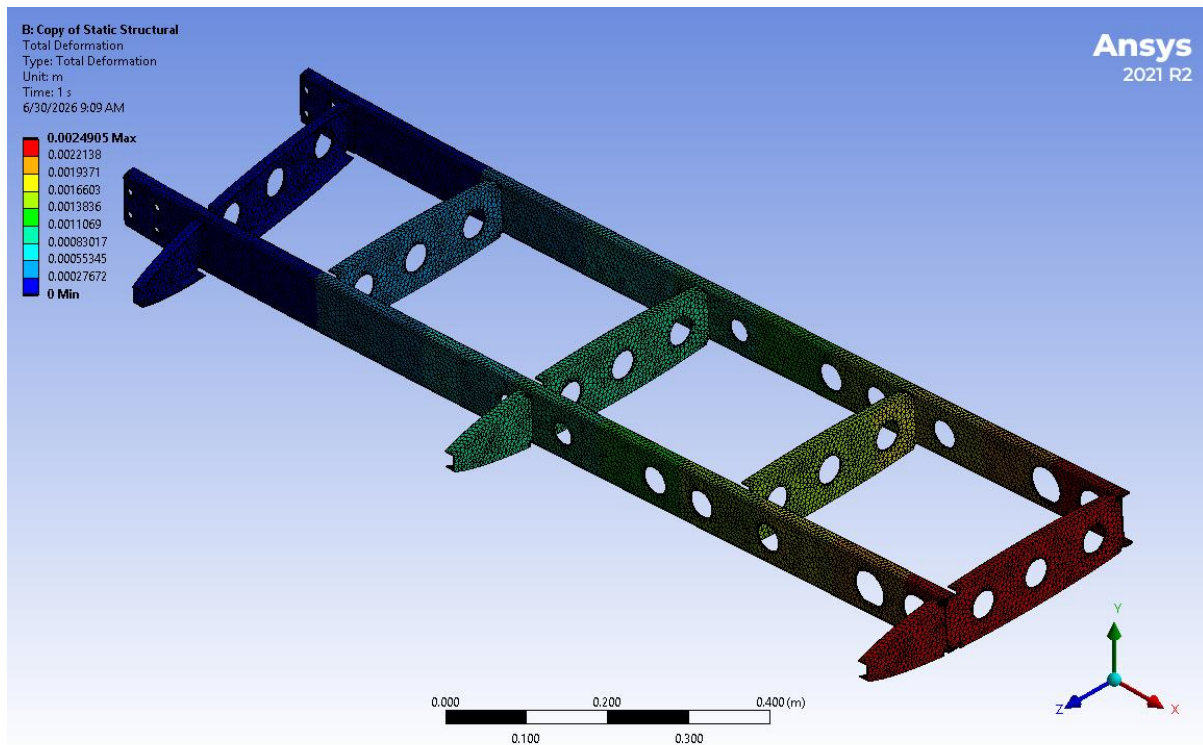


Figure 24: Total deformation variation for rear spar loaded case

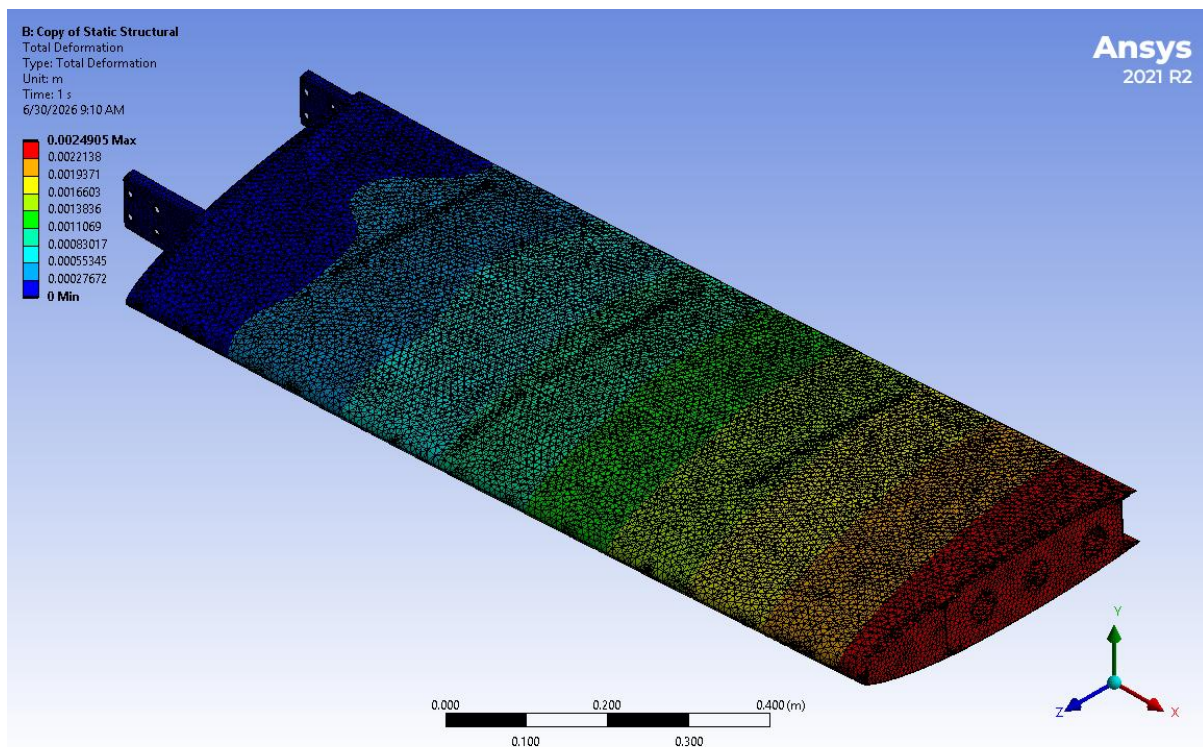


Figure 25: Total deformation variation for rear spar loaded case

4 COMPARISON OF RESULTS

The experimental and numerical results were compared for the two loading cases: load applied at the front spar tip and load applied at the rear spar tip. The experimental part was based on strain gauge readings, where the measured strains were converted into stresses using the relation $\sigma = E\varepsilon$, with $E = 69$ GPa for aluminium. The FEA part was based on the maximum deformation, maximum equivalent stress and maximum equivalent elastic strain obtained from ANSYS Static Structural.

Comparison of Experimental and FEA results

The maximum experimental stress was taken as the maximum of absolute strain gauge stress recorded in each loading case. This value is not exactly the same as the FEA maximum equivalent stress because the strain gauges measured local normal strain at selected points, while the FEA maximum equivalent stress represents the maximum value anywhere in the finite element model. Still, the comparison is useful to check whether both methods give the same order of stress and whether the trend is reasonable.

Table 9: Comparison of Experimental vs FEA results

Result	Front spar load	Rear spar load	Observation
Experimental max stress (MPa)	69.62 (at gauge B)	58.16 (at gauge B)	Rear loading reduced the peak measured stress by 16.5%
FEA max equivalent stress (MPa)	74.694	77.411	Rear loading increased FEA maximum stress by 3.6%
Deviation of FEA from Experimental results	7.29%	33.10%	Rear spar show higher deviation
Experimental max strain (micro-strain)	1009.0 (at gauge B)	842.8 (at gauge B)	Maximum measured strain was lower in rear loading
FEA max equivalent strain	1.082e-03	1.122e-03	Rear loading increased FEA maximum strain by 3.7%
FEA max deformation (mm)	2.422	2.491	Rear loading increased deformation by 2.8%

For the front spar loading case, the FEA maximum equivalent stress was 74.694 MPa, while the maximum experimental stress magnitude was 69.62 MPa. The difference is about 7.3%. For the rear spar loading case, the FEA maximum equivalent stress was 77.411 MPa and the

maximum experimental stress magnitude was 58.16 MPa, giving a higher difference of about 33.1%. This larger difference is acceptable for a preliminary comparison because the exact FEA maximum may occur near a load application point, bolt hole or local stress concentration, whereas the strain gauges did not cover every such location.

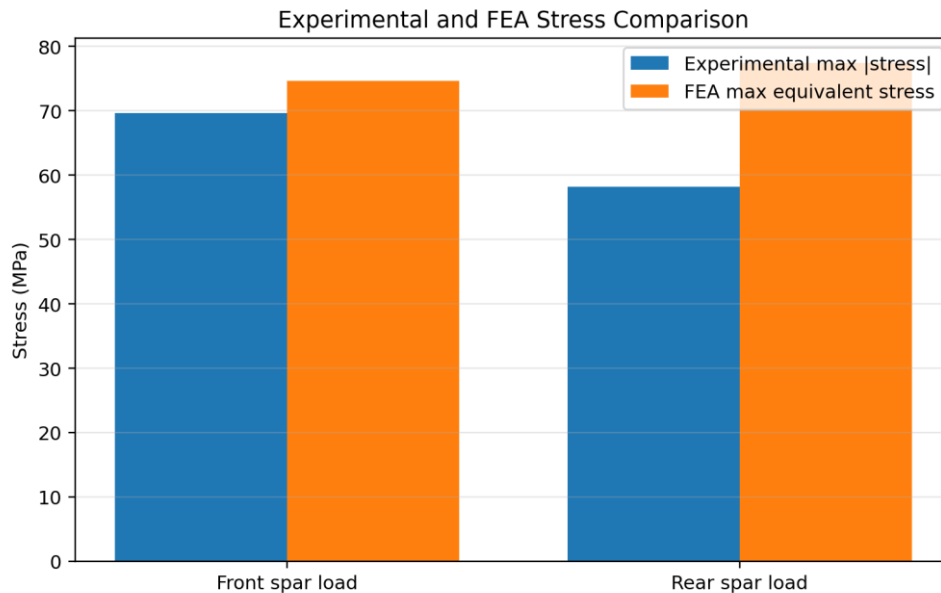


Figure 26: Comparison of Experimental and FEA results in two cases

Comparison of Spars and Skin Behaviour

Table 10: Comparison of Maximum stresses in each flange (FEA)

Structural region	Maximum stress: front load (MPa)	Maximum stress: rear load (MPa)	Main trend
Front spar flanges (A, B)	69.62	58.16	Front load higher
Front spar web (C, D, E)	12.32	14.97	Rear load higher
Upper skin (F, G, H)	67.72	33.44	Front load higher
Lower skin (I, J, K)	51.98	44.05	Front load higher
Rear spar web (L)	7.04	35.22	Rear load higher

When the load was applied to the front spar, the front spar flanges and the skin panels showed high stress magnitudes. The stress at gauge L on the rear spar web was only 7.04 MPa in magnitude. This shows that the rear spar did not carry a high direct stress in this case. When the load was applied to the rear spar, the stress at gauge L increased to 35.22 MPa and gauge K on the lower skin also increased. Therefore, the rear spar loading case transferred more load to the rear spar and lower skin. At the same time, the maximum measured stress at the front spar flange B reduced from 69.62 MPa to 58.16 MPa.

5 DISCUSSION

Selection of Landing gear mounting Spar

The experimental and numerical methods gave useful information about the structural response of the rectangular wing. The experiment directly measured strains at selected gauge positions locally while FEA showed the global deformation and maximum stress field of the complete wing model. Both methods were therefore complementary.

Based on the experimental strain gauge results, the **rear spar** is the more suitable position to mount the landing gear for this prototype wing structure. The main reason is that the maximum measured stress was lower when the load was applied to the rear spar. In addition, most strain gauge locations recorded lower stress magnitudes under rear spar loading. Therefore, the experimental load distribution is more favourable for the rear spar mounting case.

However, the FEA results should also be considered. The FEA model predicted slightly higher maximum deformation, equivalent stress and equivalent strain for rear spar loading. The increase was small, approximately 2.9% in deformation and 3.6% in equivalent stress. This indicates that the rear spar mounting position may create a local stress concentration near the load application or root constraint region. Therefore, if the rear spar is selected as the landing gear mounting location, local reinforcement or a proper bracket/load spreading fitting is recommended at the landing gear attachment point.

Assumptions made in the analysis

- ❖ The whole wing structure was assumed to be made from aluminium alloy with Young's modulus of 69 GPa.
- ❖ The wing was assumed to behave within the linear elastic region under the 30 lb applied load.
- ❖ Thermal effects, material non-linearity and plastic deformation were neglected.
- ❖ The root bolt-hole faces were modelled as fixed supports in the FEA model.
- ❖ The applied loading was represented as a point load of 30 lb, equivalent to approximately 133.45 N.
- ❖ Small geometrical details such as rivets, small fillets, some holes and non-structural details were neglected to improve mesh quality and reduce solution time.

- ❖ Contacts between structural components were assumed to be ideal, so relative slipping between members was not considered.

Limitations and possible errors

Several practical and modelling limitations may have caused differences between the experimental and FEA results. In the experiment, strain gauges measured strain only at limited locations. Therefore, local maximum stress near bolt holes, load application points or edges may not have been captured [3][4]. Strain gauge bonding quality, gauge alignment, calibration errors, bridge zero offset and small movement of the wing in the rig can also affect the measured values [3][4]. The unloaded and loaded voltage readings may also include small instrument noise.

In the finite element model, the actual bolted root condition was simplified as fixed faces. In the real setup, bolts, washers and fixture flexibility can allow slight local movement. As well as the point load representation may create local stress concentration, especially near the selected spar tip. Since the real load application may have contact area rather than a point load, the FEA maximum stress can become higher than the experimental values. Mesh density and tetrahedral element quality also affect local stress prediction [2]. Therefore, FEA maximum stress values should be interpreted carefully, especially near holes, sharp edges and load application points.

Viability of FEA results with experimental results

The FEA results were reasonably viable for a preliminary validation study. For the front spar loading case, the FEA maximum stress was only about 7.3% higher than the maximum experimental stress magnitude. This shows that the FEA model gave a close order of stress for the front spar case. For the rear spar case, the difference was higher. This does not necessarily mean the FEA was wrong, because the FEA maximum equivalent stress and experimental gauge stress are not measured at the same point and are not the same stress component. The rear spar case may also contain local stress concentration in the FEA model that was not covered by a strain gauge.

The numerical deformation results showed that rear spar loading produced slightly higher deflection than front spar loading. Since experimental deflection values were not available in the given data, this comparison was made only using the FEA results. Even though the FEA predicted a slightly higher global response for rear spar loading, the difference between the two cases was small. Therefore, the experimental stress distribution was given higher

importance when selecting the mounting location, while the FEA results were used as a warning to provide sufficient local reinforcement.

Final finding

The final decision was made by considering both load distribution and stress concentration. The experimental results showed that rear spar loading reduced the maximum measured stress and reduced the overall measured stress distribution along the gauges. This is favourable for a landing gear attachment because landing gear loads are concentrated and must be transferred into the wing box without overloading one structural member. The FEA result showed slightly higher maximum deformation and equivalent stress for rear spar loading, but the increase was small and can be controlled by local structural reinforcement. Therefore, for this prototype wing, the landing gear is recommended to be mounted on the rear spar with a properly designed attachment bracket and local reinforcement around the mounting region.

6 CONCLUSION

The structural behaviour of the rectangular wing was analysed experimentally and numerically for two landing gear load positions. In the experimental study, the highest stress for both loading cases occurred at gauge B on the lower flange of the front spar. However, the maximum measured stress was reduced from 69.62 MPa to 58.16 MPa when the load was moved from the front spar to the rear spar. Most of the strain gauge readings also showed lower stress magnitudes in the rear spar loading case. Therefore, the experimental results indicate that rear spar loading gives a more favourable load distribution.

In the FEA study, the rear spar loading case showed slightly higher maximum deformation, equivalent stress and equivalent strain compared with the front spar loading case. The increase was small, but it shows that local reinforcement is needed if the landing gear is mounted on the rear spar. Considering both experimental and numerical results, the rear spar is recommended as the better landing gear mounting location for the prototype wing, provided that a suitable load-spreading attachment and reinforcement are used.

7 REFERENCES

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